Seed Dispersal Derby

Build a seed that escapes the parent tree without engines or throwing.

Win condition: Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.

THE SCIENCE YOU ARE USING

Drag, lift, and hang time. A maple samara spins as it falls. The spinning wing pushes against the air and makes lift, so the seed drifts down slowly and even a light breeze carries it away from the shade of its parent. More surface area for the same mass usually means a longer, slower fall.

One variable at a time. Wing length, fold angle, and added mass all change flight. Engineers learn what each does by changing only one and comparing. Copying a natural strategy on purpose is biomimicry.

HOW TO BUILD AND RUN IT

01 **Study a real samara.** Drop a maple seed or a model. Watch how the wing makes it spin and slow down.

02 **Build a baseline seed.** Make a simple winged seed with one paper clip for mass. Record a baseline drop time and distance.

03 **Change one variable.** Adjust wing length OR mass OR fold angle, only one. Predict the effect before testing.

04 **Test for hang time and distance.** Drop from the standard height or release in the fan lane. Time three trials and average.

05 **Aim for a target.** In a final round, try to land in a marked zone. Control plus distance both score.

Safety: No throwing at people. Use marked launch lanes and low-power fan settings. Allergy: None.

MATERIALS

Paper and cardstock	shared, for the wing
Paper clips	for mass, a few per team
Masking tape	shared
String	shared, to mark the drop height
Fan or marked drop lane	1 shared test zone (low setting)
Tape measure	1 shared, to measure distance
Timers	1 per team (4 teams)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________________

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The ONE variable we are testing: ______________________________

Baseline (average of 3 drops): hang time ______ **s, distance** ______ **cm**

After redesign (average of 3 drops): hang time ______ **s, distance** ______ **cm**

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________________

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If you ran it again, what is the ONE thing you would change? ______________________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**